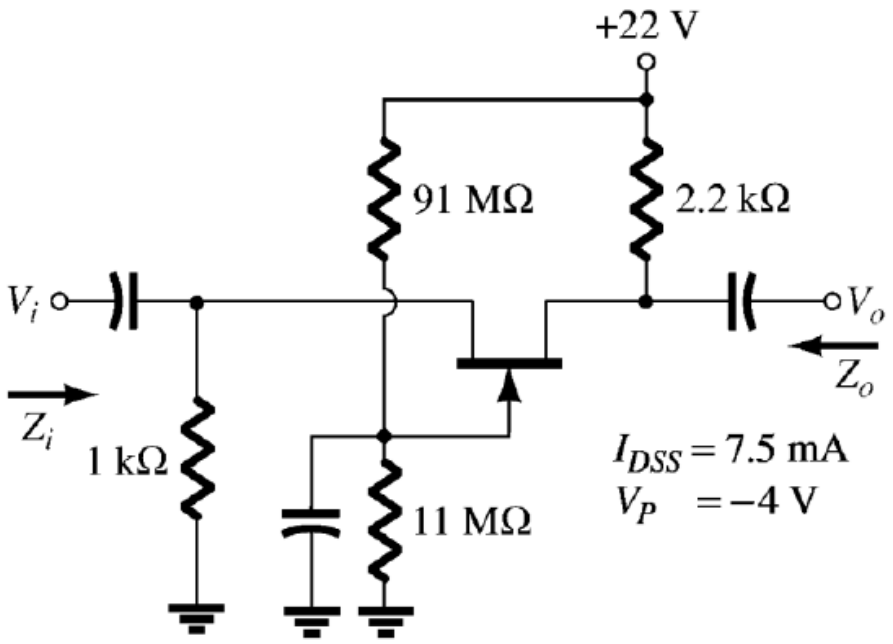
FET small-signal II

Practice questions · Day 20 · Thu 12 Nov · pack E12

5. Practice

From your course files

WS9 Q5 (course question). Determine Z_i, Z_o, and A_v for the network of Fig. 5 if $r_d = 30$ kΩ.

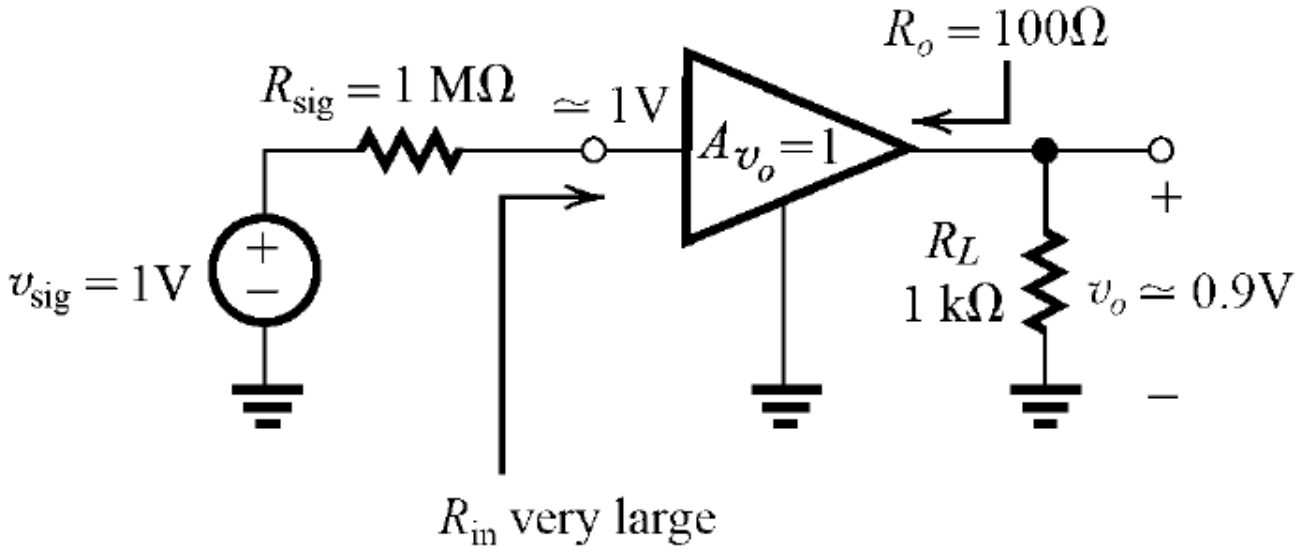


WS9 Fig. 5 (from the workshop sheet)

WS9 Q6 (course question). A common-gate (CG) amplifier is required to match a signal source with $R_{sig} = 100\ \Omega$. At what current should the MOSFET be biased if it is operated at an overdrive voltage of 0.20 V? If the total resistance in the drain circuit is 2 kΩ, what overall voltage gain is realized?

WS9 Q7 (course question). It is required to design a source follower that implements the buffer amplifier shown in Fig. 6. If the MOSFET is operated with an overdrive voltage

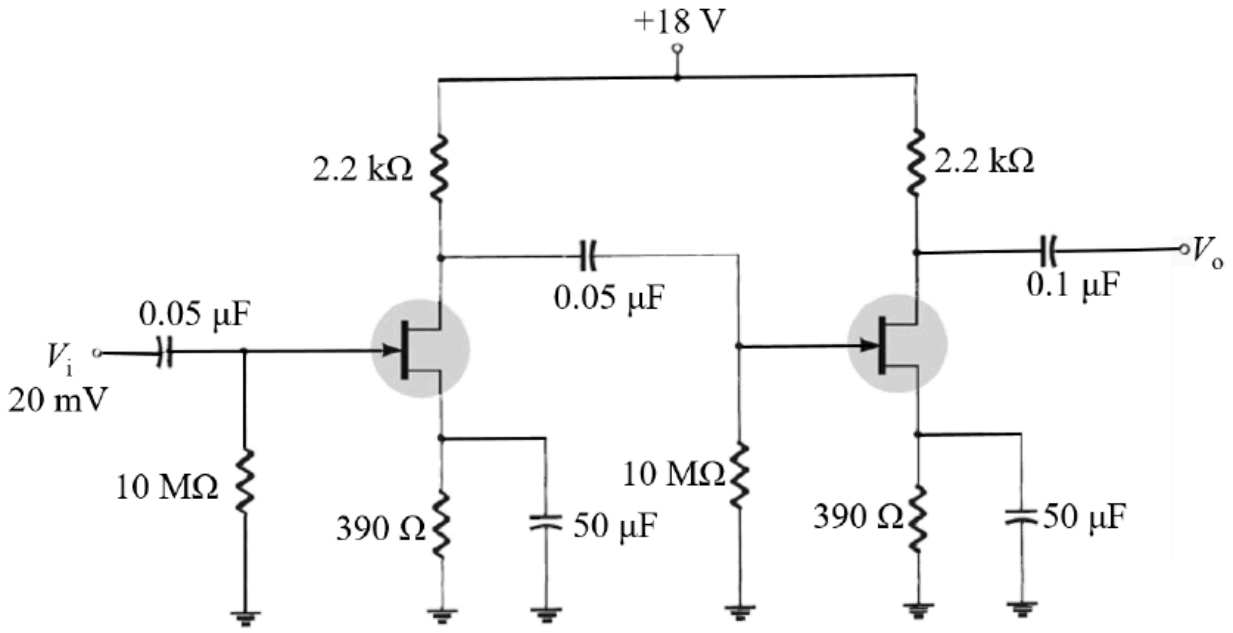
$V_{ov} = 0.25$ V, at what drain current should it be biased? Find the output signal amplitude and the signal amplitude between gate and source.



WS9 Fig. 6 (from the workshop sheet)

WS9 Q8 (course question). A MOSFET is connected in the source-follower configuration and employed as the output stage of a cascade amplifier. It is required to provide an output resistance of 200 Ω. If the MOSFET has $k = 0.4$ mA/V^2 and is operated at $V_{ov} = 0.25$ V, find the required W/L ratio. Also specify the dc bias current If the amplifier load resistance varies over the range 1 kΩ to 10 kΩ what is the range of G_v of the source follower?

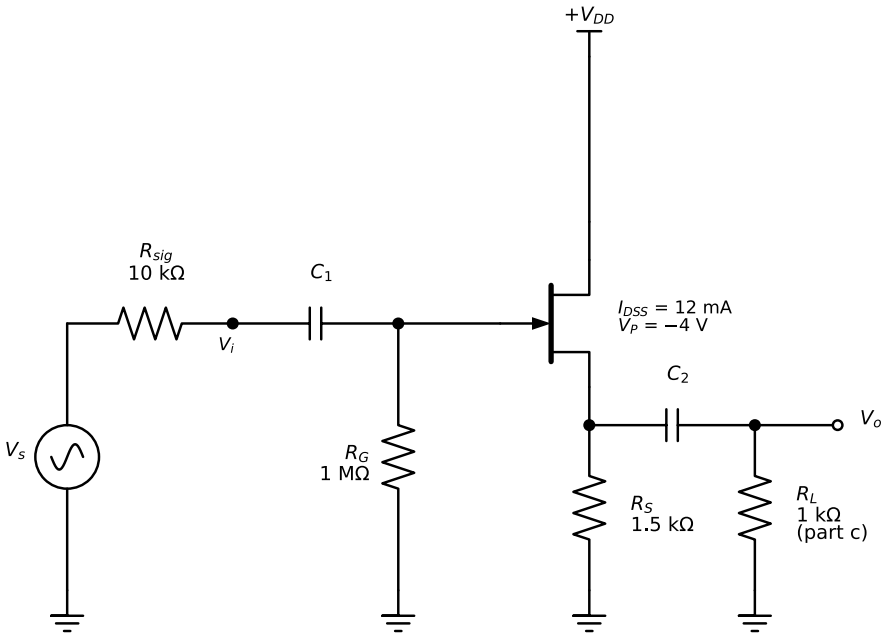
WS9 Q9 (course question). For the JFET cascade amplifier of Fig. 7, using identical JFETs with $I_{DSS} = 8$ mA and $V_P = -4.5$ V, calculate the voltage gain of each stage, the overall gain of the amplifier, and the output voltage V_o.



WS9 Fig. 7 (from the workshop sheet)

New questions

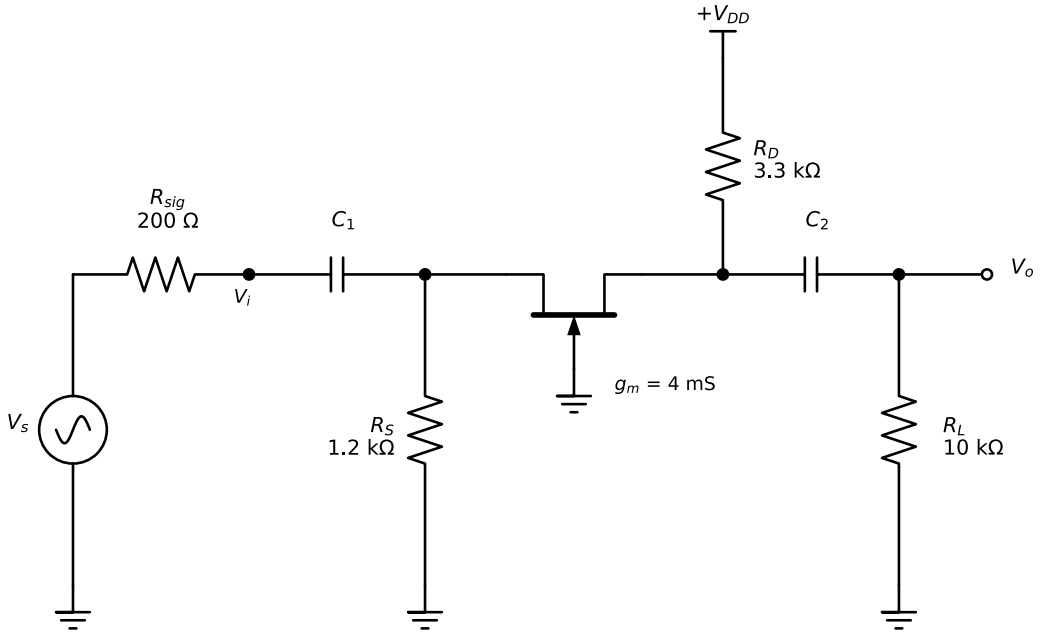
N1. [3 marks] An n-channel E-MOSFET has $k = 0.3$ mA/V² and $V_{GS(\mathrm{Th})} = 2$ V, biased at $V_{GSQ} = 5$ V. Find I_{DQ} and g_m, and confirm g_m with the overdrive form.



Circuit for N2: JFET source follower with source resistance and load

N2. [7 marks] A JFET source follower has $I_{DSS} = 12$ mA, $V_P = -4$ V, $R_G = 1$ MΩ (gate to ground), $R_S = 1.5$ kΩ (source to ground), drain to V_{DD}, $r_d = \infty$.

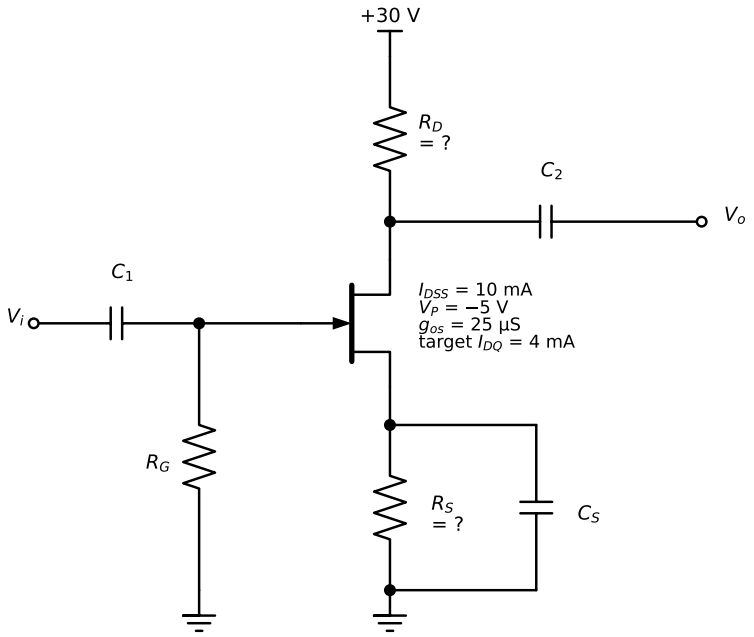
a. Find I_{DQ}, V_{GSQ} and g_m. [3]
b. Find Z_i, Z_o and the no-load A_v. [2]
c. The stage drives $R_L = 1$ kΩ from a source with $R_{sig} = 10$ kΩ. Find A_{v_L} and A_{v_s}. [2]



Circuit for N3: common-gate stage fed from R_sig and driving R_L

N3. [6 marks] A common-gate stage has $g_m = 4$ mS, $R_S = 1.2$ kΩ, $R_D = 3.3$ kΩ, $r_d = \infty$. It is fed from $R_{sig} = 200\ \Omega$ and drives $R_L = 10$ kΩ.

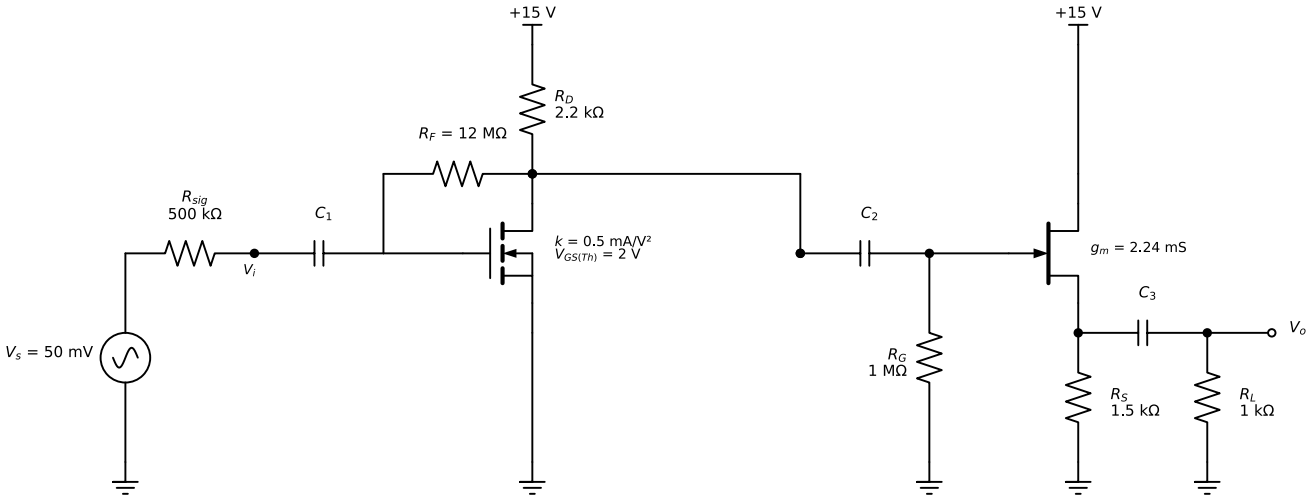
a. Find Z_i and Z_o. [2]
b. Find the no-load gain, A_{v_L} and A_{v_s}. [3]
c. In one sentence, why is the overall gain about half of A_{v_L}? [1]



Circuit for N4: self-bias JFET stage to be designed

N4. [8 marks] Design a self-bias JFET stage with R_S fully bypassed. $I_{DSS} = 10$ mA, $V_P = -5$ V, $g_{os} = 25$ µS, $V_{DD} = 30$ V, target $I_{DQ} = 4$ mA and $A_v = -10$.

a. Find V_{GSQ} and R_S. [2]
b. Find g_m. [1]
c. Find R_D for $A_v = -10$, then the gain with the standard values 4.3 kΩ and 4.7 kΩ. [3]
d. Check the FET stays in saturation with the exact R_D. [1]
e. If C_S is removed, what is A_v (exact R_D, include r_d)? [1]



Circuit for N5: E-MOSFET drain-feedback stage driving a JFET source follower

N5. [10 marks] Two-stage cascade. Stage 1 is an E-MOSFET drain-feedback amplifier: $V_{DD} = 15$ V, $R_D = 2.2$ kΩ, $R_F = 12$ MΩ, $k = 0.5$ mA/V², $V_{GS(\text{Th})} = 2$ V. Stage 2 is the source follower of N2 ($R_G = 1$ MΩ, $g_m = 2.24$ mS, $R_S = 1.5$ kΩ), driving $R_L = 1$ kΩ. The source is $V_s = 50$ mV with $R_{sig} = 500$ kΩ. $r_d = \infty$ for both.

a. Find I_{DQ}, V_{GSQ} and g_m for stage 1. [3]
b. Find A_{v1} (loaded by stage 2) and Z_{i1}. [3]
c. Find A_{v2}, the overall $A_v = V_o/V_i$, and V_o. [3]
d. State Z_o of the cascade. [1]

N6. [5 marks] (Sedra form.) An NMOS has $\mu_n C_{ox} = 0.2$ mA/V², $W/L = 40$, $V_t = 0.6$ V and $I_D = 0.5$ mA.

a. Find V_{ov}, V_{GS} and g_m. [3]
b. Give the course k and confirm $g_m = 2kV_{ov}$. [1]
c. Find the CS gain with $R_D = 5$ kΩ and $R_L = 10$ kΩ. [1]

6. Quick recall check

1. Write Z_o for a source follower with $r_d = \infty$.
2. What is Z_i (approximate) for an E-MOSFET drain-feedback amplifier?
3. A CG stage has $g_m = 5$ mS. What is its approximate Z_i if $R_S \gg 200\ \Omega$?
4. In Sedra notation, $I_D = 2$ mA and $V_{ov} = 0.4$ V. What is g_m?
5. Two identical CS stages each have $A_v = -4$. What is the overall gain, and is the output in phase with the input?